

Modeling Dynamics: Autoregressive and Cross-Lagged Panel Models

Day Two

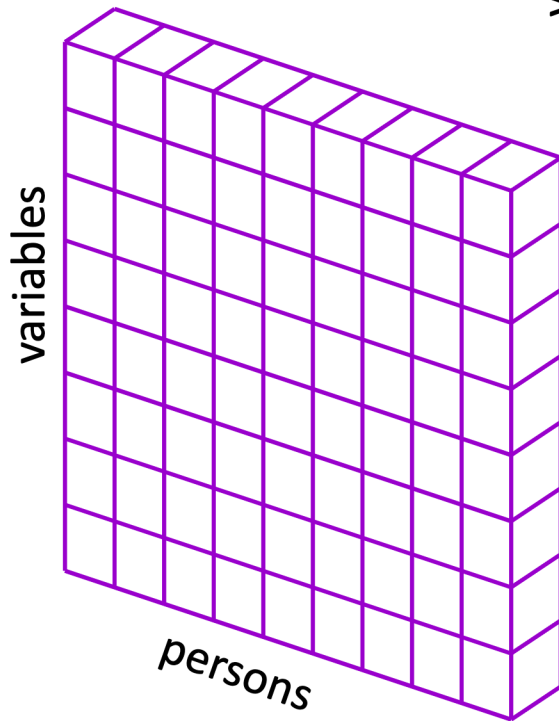
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Assistant Professor in Quantitative Methodology
Department of Population Health Sciences

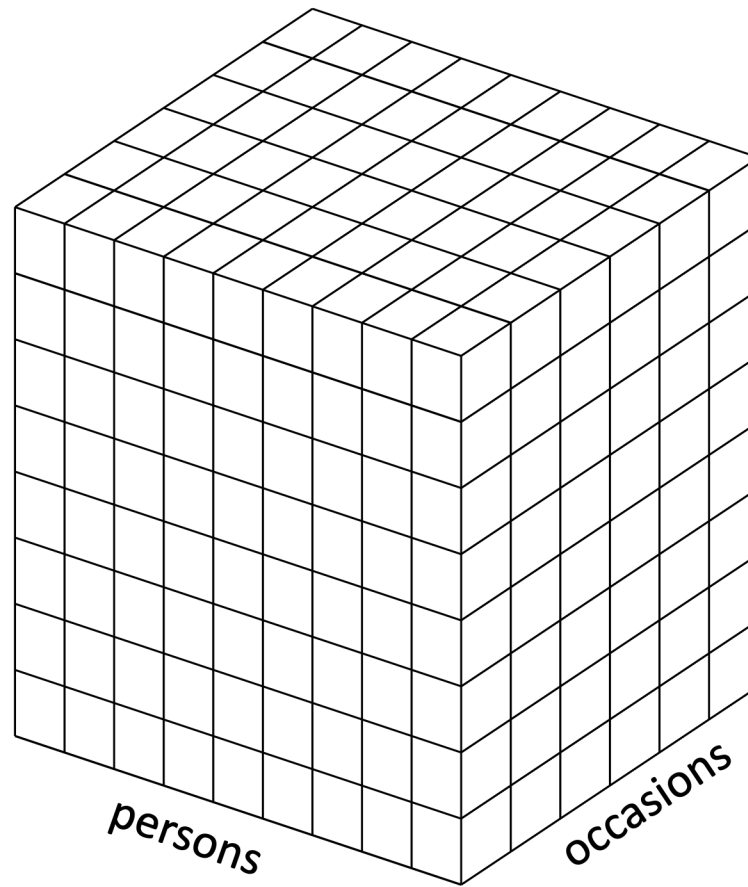
Overview

- $N > 1$ data
- Multilevel autoregressive model
- Panel data
- Cross-lagged panel model
- Random-intercept cross-lagged panel model

**Inter-individual or
cross-sectional data**



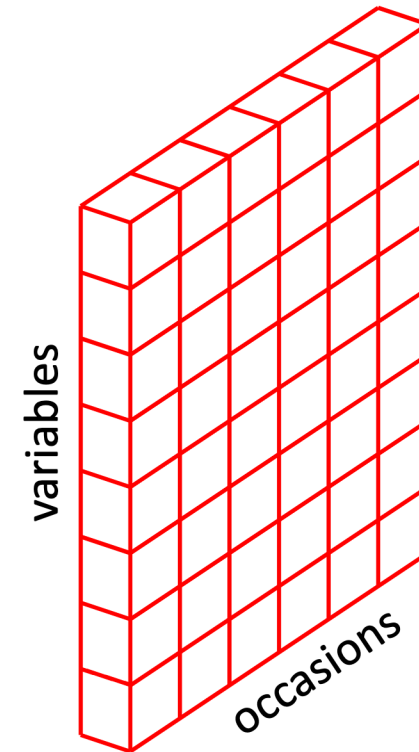
variables



persons

occasions

**Intra-individual or
time series data**



variables

occasions

Cattell's data box

From Hamaker, E. L. (2025). Common data types: Cross-sectional, time series, panel, and intensive longitudinal data. *MATILDA Preprints*, 2025-05-23.

<https://matilda.fss.uu.nl/articles/common-data-types.html>

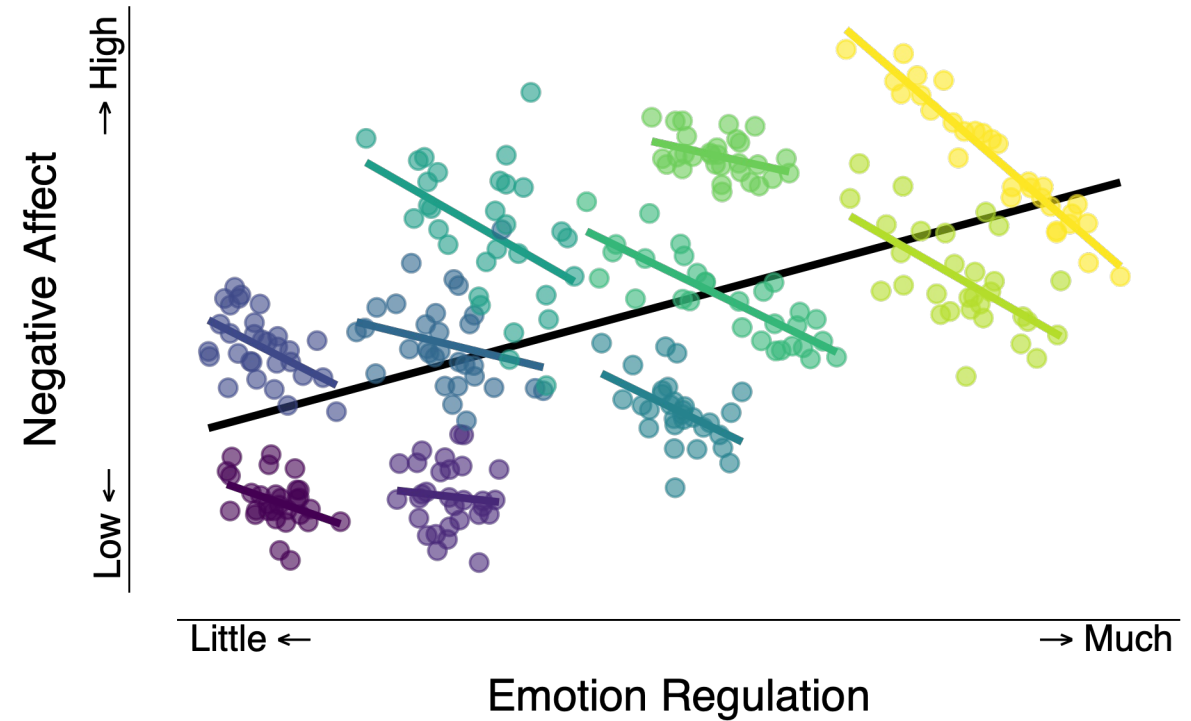
Multilevel autoregressive modeling

Within-person effects are not necessarily similar to between-person effects

Simpson's paradox

People with higher negative affect tend to regulate their emotions more than others.

However, when someone regulates their emotions more, their negative affect tends to be lower than when they regulate less.



N > 1 ILD

Allows you to estimate both within- and between person effects.

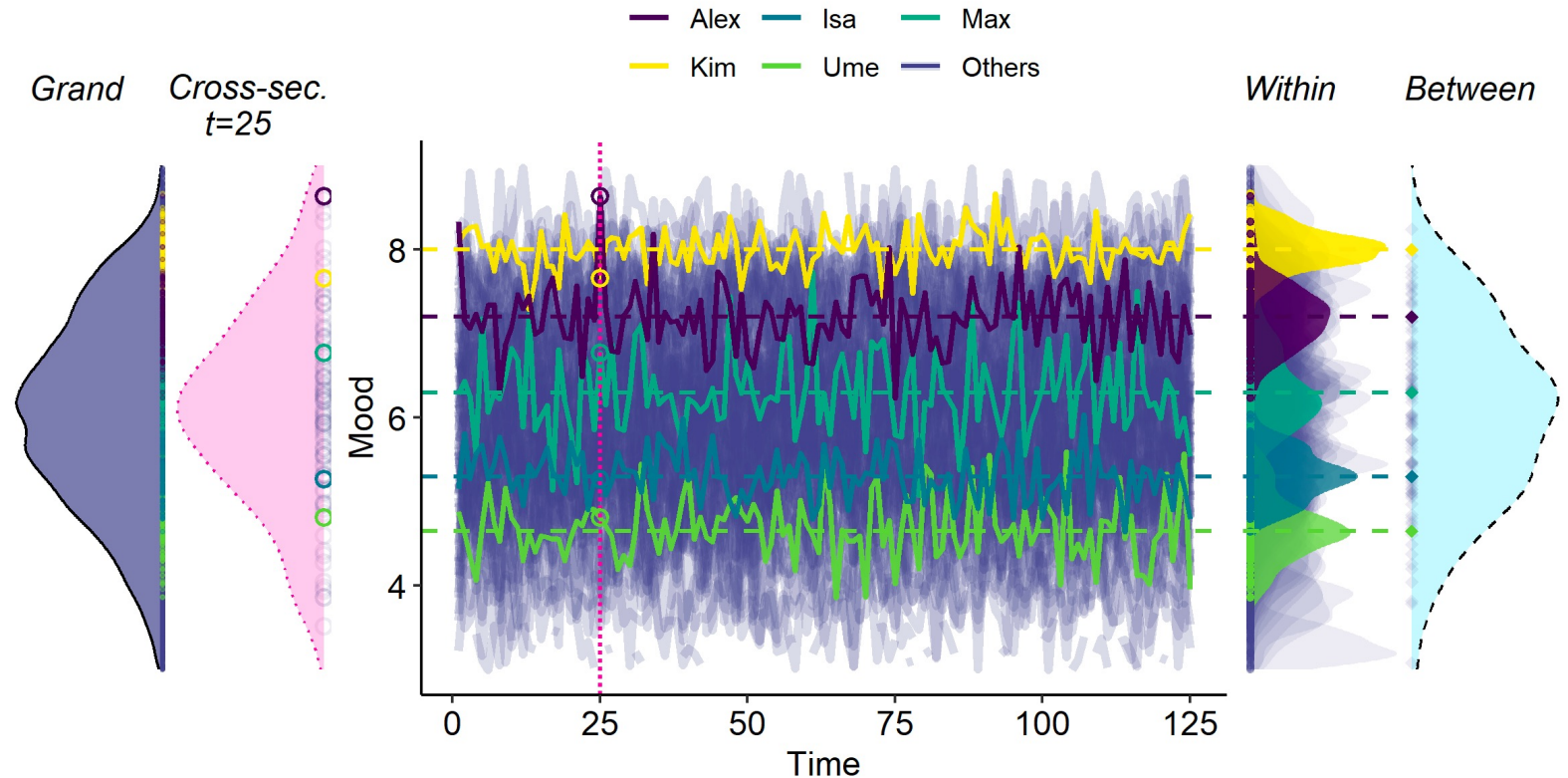
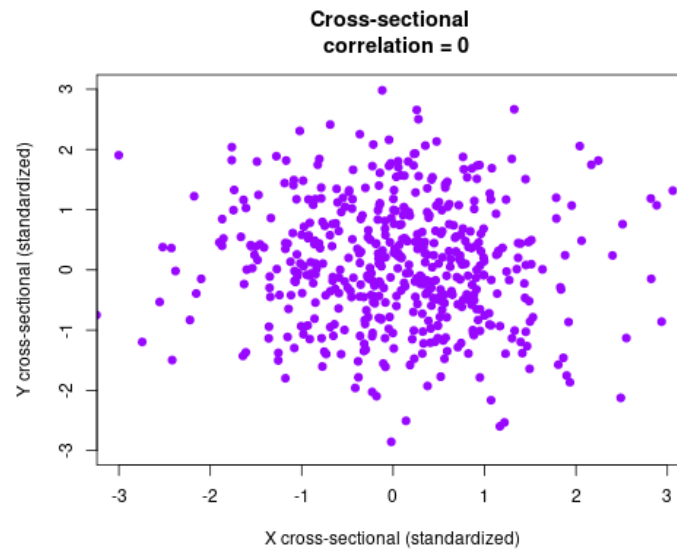
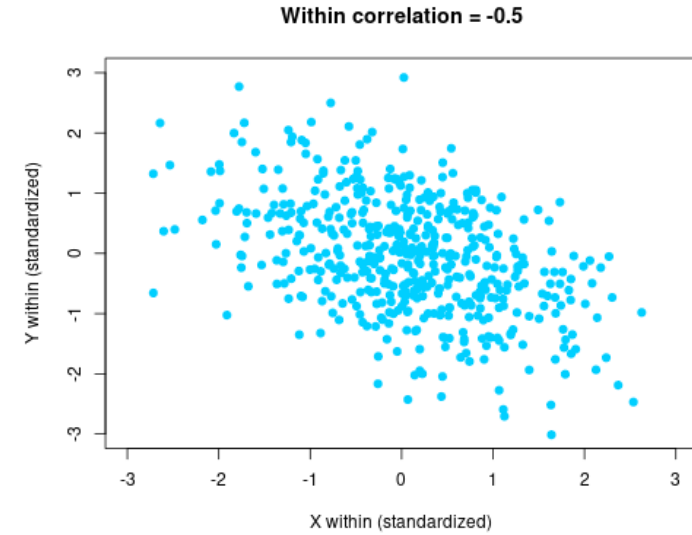
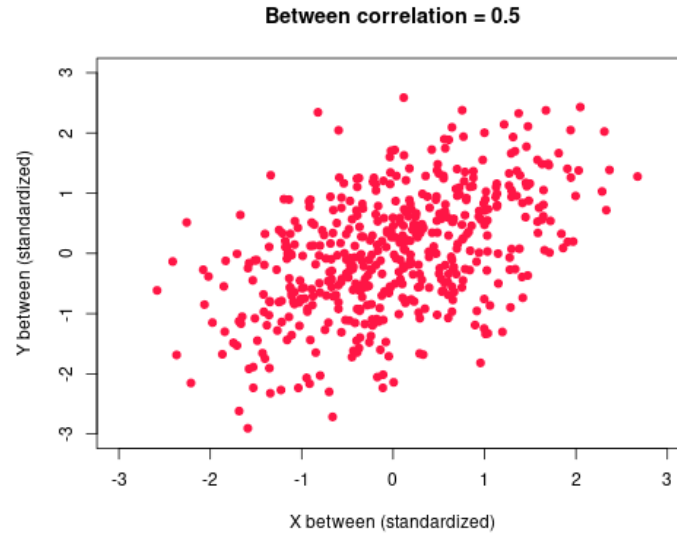
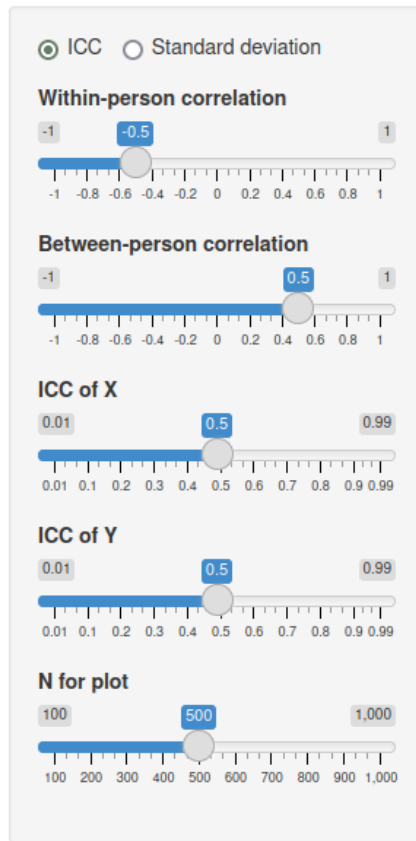


Figure from Schuurman, N. (2023). A "Within/Between Problem" Primer: About (not) separating within-person variance and between-person variance in psychology. PsyArXiv. <https://doi.org/10.31234/osf.io/7zgkx>



A Within/Between Problem App

From Schuurman, N. (2023). A "Within/Between Problem" Primer: About (not) separating within-person variance and between-person variance in psychology. PsyArXiv.

<https://doi.org/10.31234/osf.io/7zgkx>

Multilevel AR(1)

Within-person autoregressive process.

Between-person differences in the mean, autoregression, and innovation variance.

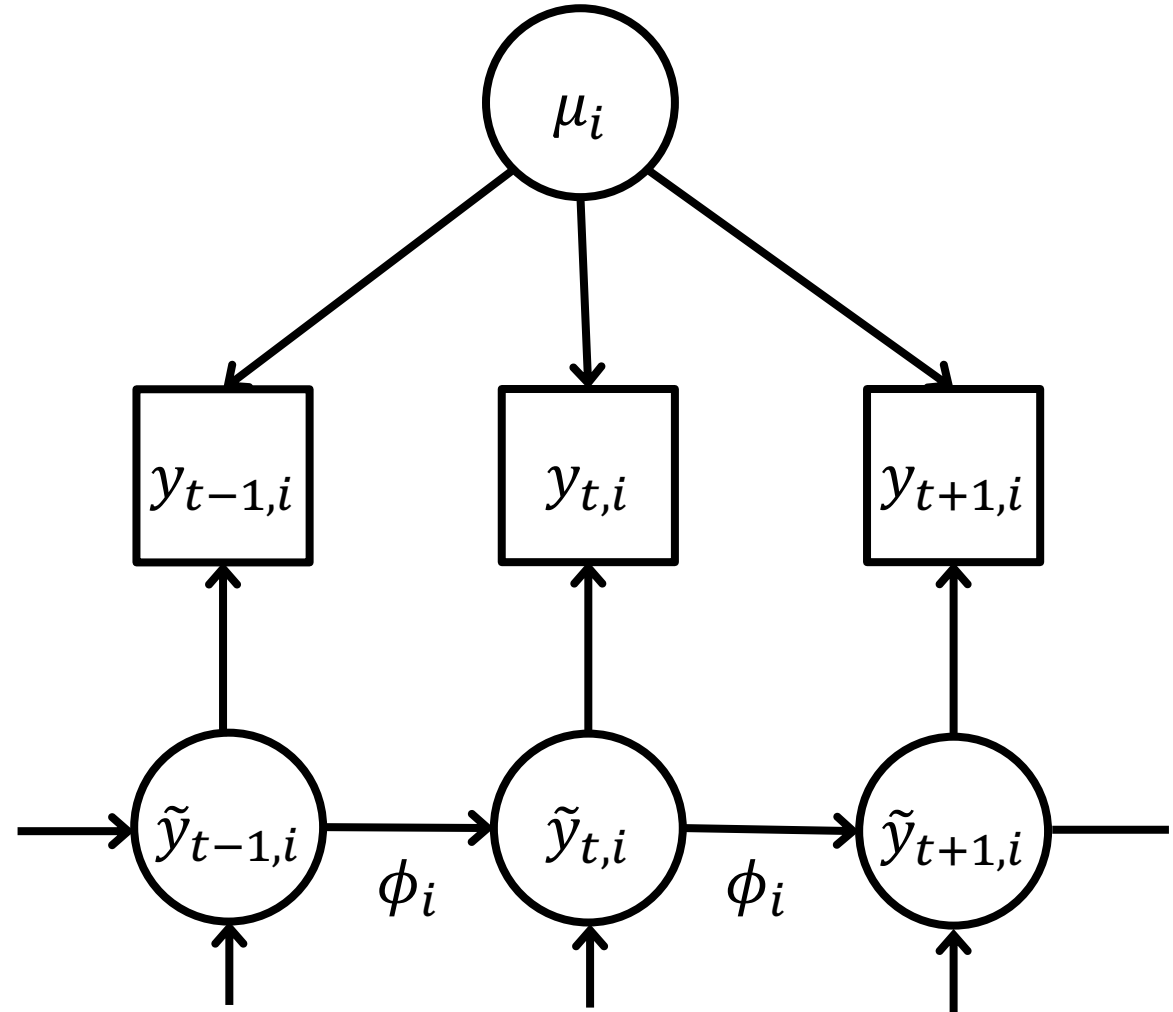
Multilevel AR(1)

$$y_{t,i} = \mu_i + \tilde{y}_{t,i}$$

$$\tilde{y}_{t,i} = \phi_i \cdot \tilde{y}_{t-1,i} + \varepsilon_{t,i}$$

$$\varepsilon_{t,i} \sim \mathcal{N}(0, \psi_i)$$

$$\begin{bmatrix} \mu_i \\ \phi_i \\ \psi_i \end{bmatrix} = \mathcal{N} \left(\begin{bmatrix} \gamma_\mu \\ \gamma_\phi \\ \gamma_\psi \end{bmatrix}, \begin{bmatrix} v_\mu^2 & & \\ v_{\mu,\phi} & v_\phi^2 & \\ v_{\mu,\psi} & v_{\phi,\psi} & v_\psi^2 \end{bmatrix} \right)$$



Multilevel AR(1) DSEM graph

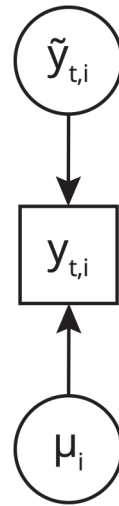
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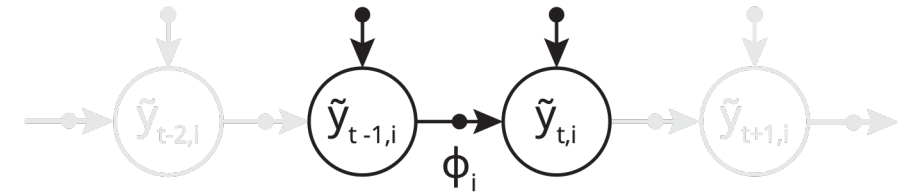
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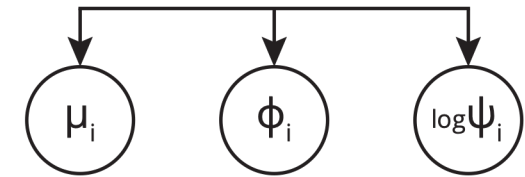
Decomposition



Within level



Between level

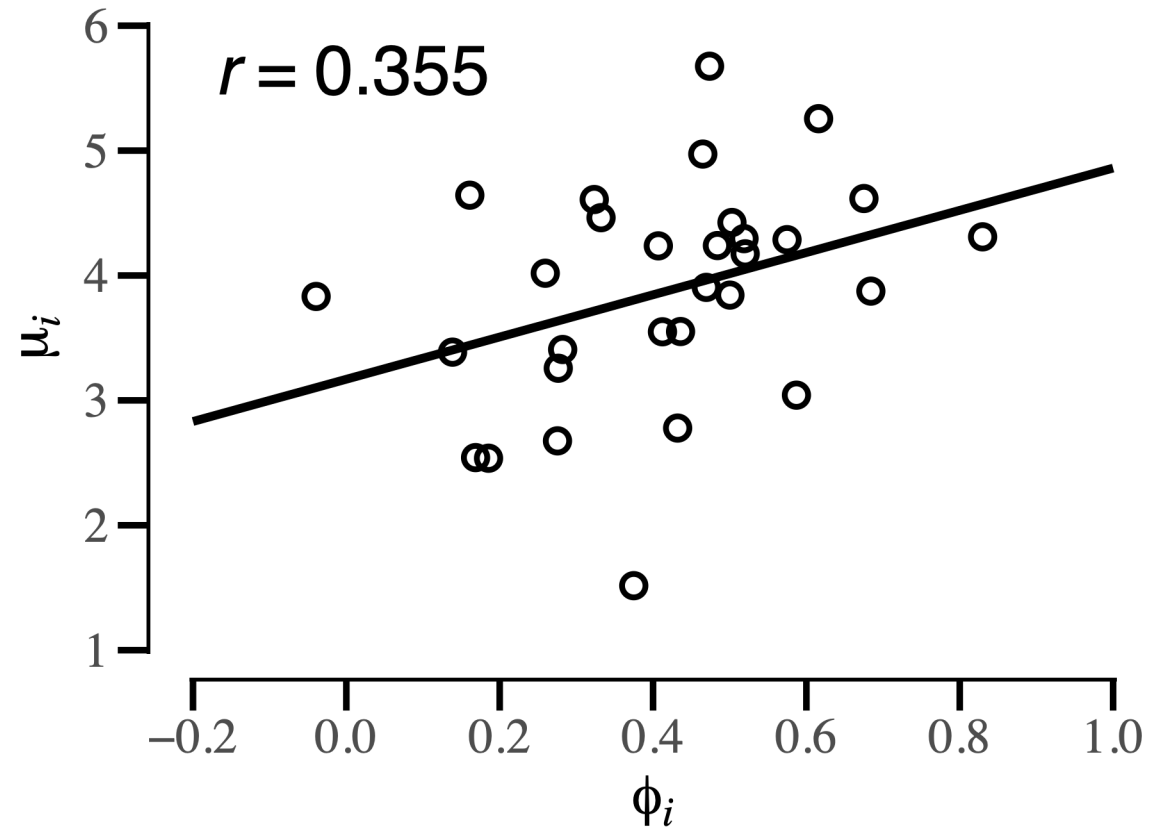


Random effects

Correlations between random effects may be of interest.

e.g., do people with higher average negative affect tend to have a higher autoregression?

(but see Dejonckheere, E., Mestdagh, M., Houben, M., Rutten, I., Sels, L., Kuppens, P., & Tuerlinckx, F. (2019). Complex affect dynamics add limited information to the prediction of psychological well-being. *Nature Human Behaviour*, 3(5), 478–491. <https://doi.org/10.1038/s41562-019-0555-0>)



Random effects

You may also be interested to see whether certain between-person differences can predict random effects.

e.g., do people who score higher on neuroticism have higher emotional inertia?

Data format

Be careful when creating a lagged variable!

ID	t	$y_{t,i}$	y_{t-1}
1	1	$y_{1,1}$	NA
1	2	$y_{2,1}$	$y_{1,1}$
1	3	$y_{3,1}$	$y_{2,1}$
1	4	$y_{4,1}$	$y_{3,1}$
1	5	$y_{5,1}$	$y_{4,1}$
1	T	$y_{T,1}$	$y_{T-1,1}$
2	1	$y_{1,2}$	$y_{T,1}$
2	2	$y_{2,2}$	$y_{1,2}$
2	3	$y_{3,2}$	$y_{2,2}$
2	4	$y_{4,2}$	$y_{3,2}$
2	5	$y_{5,2}$	$y_{4,2}$
2	T	$y_{T,2}$	$y_{T-1,2}$

Should be NA!

Multilevel AR(1) estimation

Well established in Mplus:

Dynamic Structural Equation Modeling (DSEM)

In R:

Univariate estimation with the **lme4** package

Implemented in the **mlVAR** or **psychonetrics** packages

New package **mlts** offers more DSEM functionality

Multilevel AR(1) exercises

Fit a multilevel AR(1) model to empirical data.

Open `multilevel_ar.Rmd`

N > 1 panel data

Usually in wide format

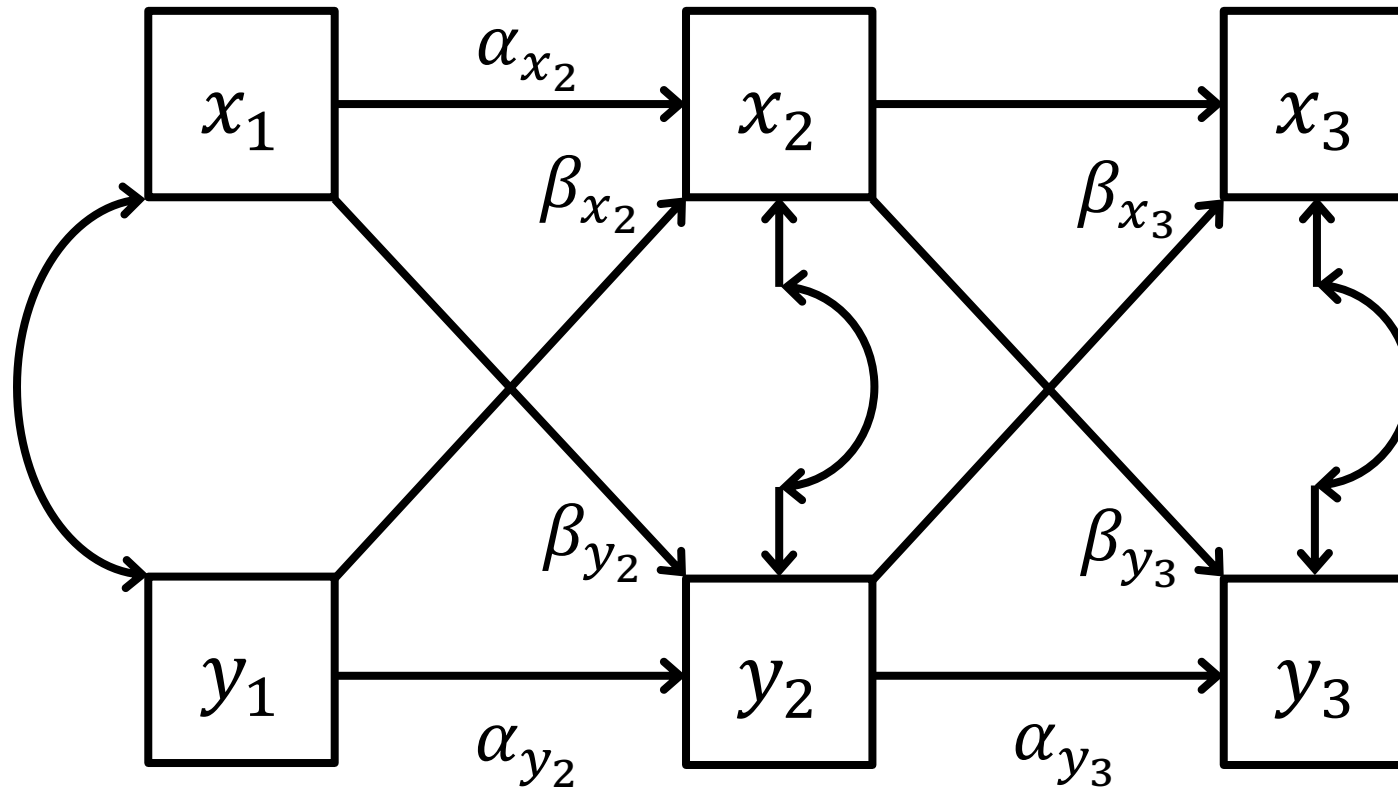
ID	$y_{1,i}$	$y_{2,i}$	$y_{3,i}$
1	$y_{1,1}$	$y_{2,1}$	$y_{3,1}$
2	$y_{1,2}$	$y_{2,2}$	$y_{3,2}$
3	$y_{1,3}$	$y_{2,3}$	$y_{3,3}$

The cross-lagged panel model

Cross-lagged panel model (CLPM)

$$x_t = c_{x,t} + \alpha_{x,t-1} \cdot x_{t-1} + \beta_{x,t-1} \cdot y_{t-1} + v_{x,t}$$

$$y_t = c_{y,t} + \alpha_{y,t-1} \cdot y_{t-1} + \beta_{y,t-1} \cdot x_{t-1} + v_{y,t}$$



Examples

The reciprocal relation between positive and negative affect within a person.

The reciprocal relation between negative affect within a dyad (e.g., a romantic couple or a parent and a child).

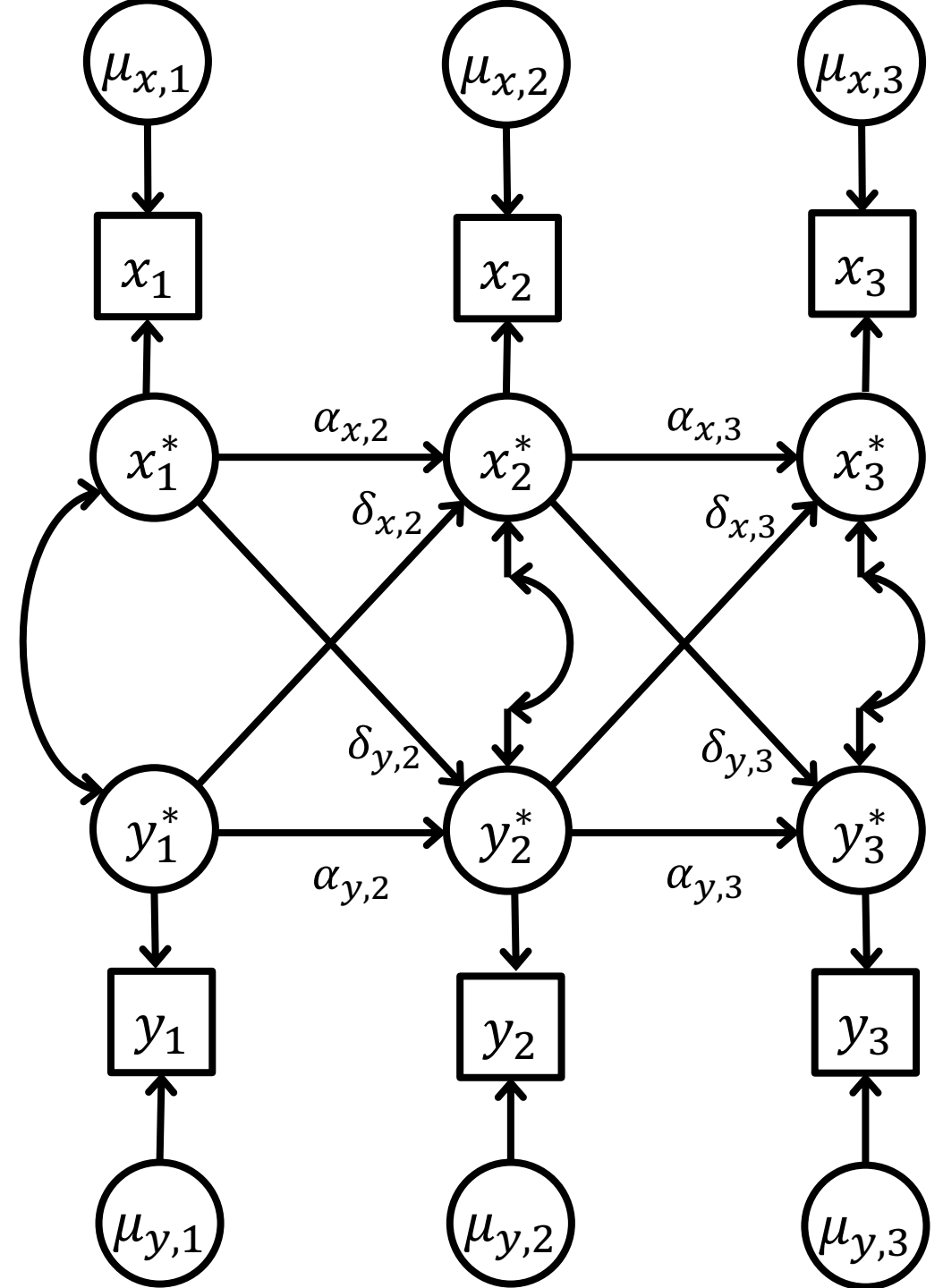
CLPM alternative formulation

$$x_t = \mu_{x,t} + x_t^*$$

$$x_t^* = \alpha_{x,t-1} \cdot x_{t-1}^* + \delta_{x,t-1} \cdot y_{t-1}^* + v_{x,t}$$

$$y_t = \mu_{y,t} + y_t^*$$

$$y_t^* = \alpha_{y,t-1} \cdot y_{t-1}^* + \delta_{y,t-1} \cdot x_{t-1}^* + v_{y,t}$$



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$$y_t = \mu_{y,t} + y_t^*$$

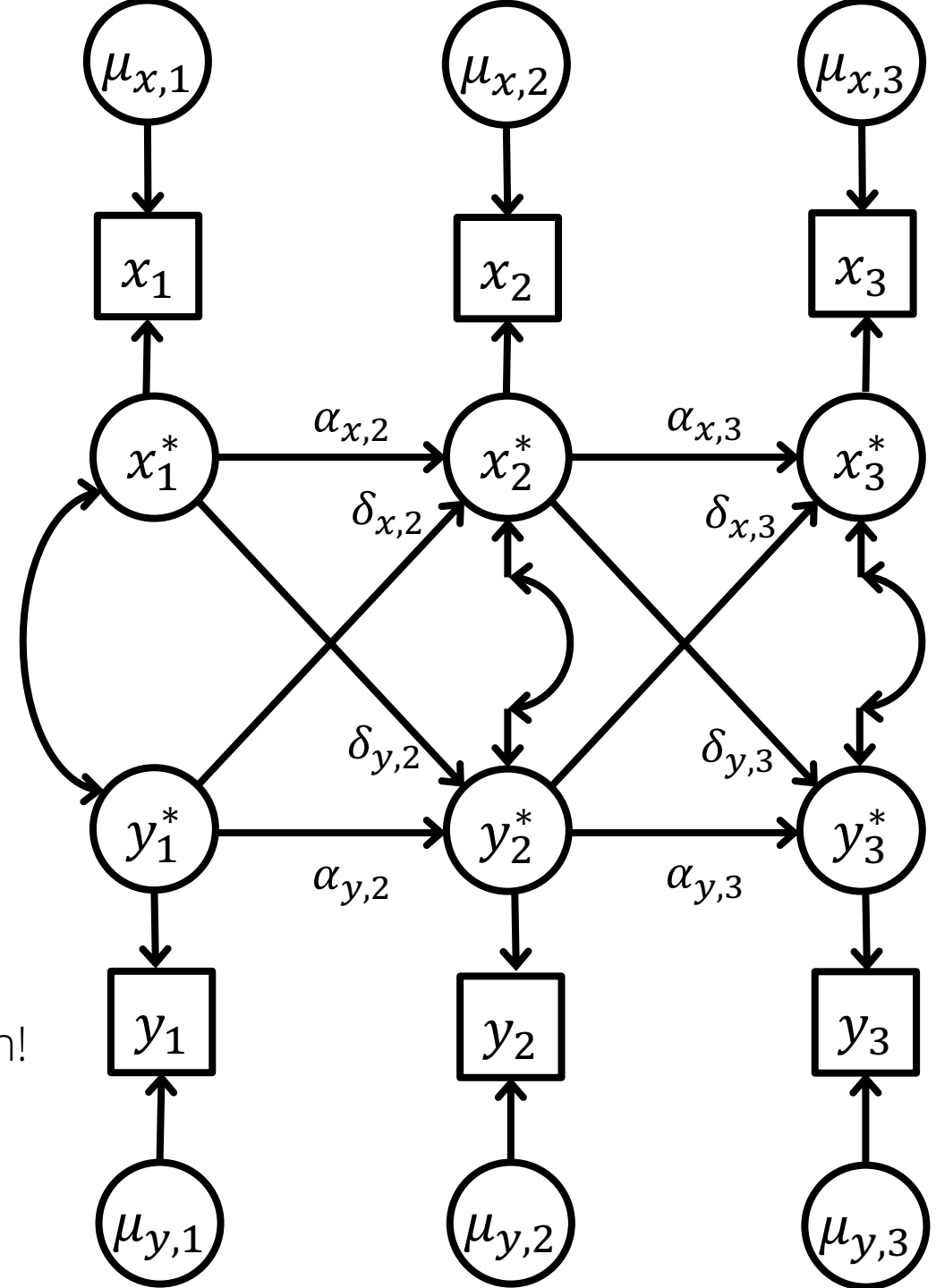
$$y_t^* = \alpha_{y,t-1} \cdot y_{t-1}^* + \delta_{y,t-1} \cdot x_{t-1}^* + v_{y,t}$$

Note that $\mu_{x,t}$ and $\mu_{y,t}$ are grand means.

x_t^* and y_t^* are individual deviations from the means.

→ Everybody fluctuates around the same mean.

Recall: The mean matters in estimating autoregression!



CLPM alternative formulation

$$x_t = \mu_{x,t} + x_t^*$$

$$x_t^* = \alpha_{x,t-1} \cdot x_{t-1}^* + \delta_{x,t-1} \cdot y_{t-1}^* + v_{x,t}$$

$$y_t = \mu_{y,t} + y_t^*$$

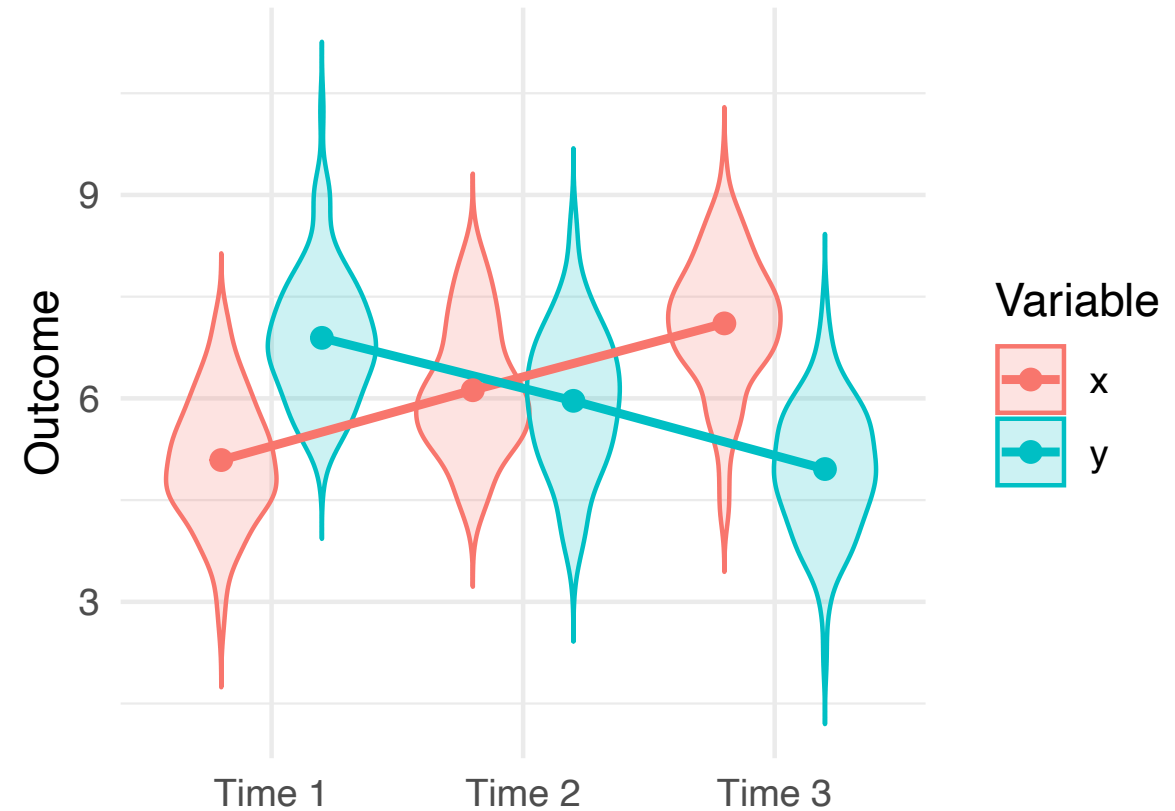
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CLPM interpretation

Autoregressive effects α

Included for stability of the construct.

Interpreted as temporal stability.

Cross-lagged effects δ

Used to study reciprocal causal effects.

Interpreted as predicting change.

Standardized to determine causal dominance.

Cross-lagged effects δ : Granger causality

If x precedes y

→ x is a 'Granger-cause' of y

... but actually means x predicts y

Estimation in R

Using the **lavaan** package, model specification is in **lavaan** code and offers a lot of flexibility.

The **psychonetrics** package also offers CLPM estimation through the **panelvar()** function, but is less flexible.

CLPM exercises

Fit a CLPM to empirical data

Open clpm.Rmd

But...

The CLPM cannot account for between-person differences.

→ The lagged parameters do not represent actual within-person relationships.

A Critique of the Cross-Lagged Panel Model

Ellen L. Hamaker and Rebecca M. Kuiper
Utrecht University

Raoul P. P. P. Grasman
University of Amsterdam

The cross-lagged panel model (CLPM) is believed by many to overcome the problems associated with the use of cross-lagged correlations as a way to study causal influences in longitudinal panel data. The current article, however, shows that if stability of constructs is to some extent of a trait-like, time-invariant nature, the autoregressive relationships of the CLPM fail to adequately account for this. As a result, the lagged parameters that are obtained with the CLPM do not represent the actual within-person relationships over time, and this may lead to erroneous conclusions regarding the *presence*, *predominance*, and *sign* of causal influences. In this article we present an alternative model that separates the within-person process from stable between-person differences through the inclusion of random intercepts, and we discuss how this model is related to existing structural equation models that include cross-lagged relationships. We derive the analytical relationship between the cross-lagged parameters from the CLPM and the alternative model, and use simulations to demonstrate the spurious results that may arise when using the CLPM to analyze data that include stable, trait-like individual differences. We also present a modeling strategy to avoid this pitfall and illustrate this using an empirical data set. The implications for both existing and future cross-lagged panel research are discussed.

Keywords: cross-lagged panel, reciprocal effects, longitudinal model, trait-state models, within-person dynamics

But...

The CLPM cannot account for between-person differences.

→ The lagged parameters do not represent actual within-person relationships.

General Article

Why the Cross-Lagged Panel Model Is Almost Never the Right Choice

Richard E. Lucas 

Department of Psychology, Michigan State University, East Lansing, Michigan

Abstract

The cross-lagged panel model (CLPM) is a widely used technique for examining reciprocal causal effects using longitudinal data. Critics of the CLPM have noted that by failing to account for certain person-level associations, estimates of these causal effects can be biased. Because of this, models that incorporate stable-trait components (e.g., the random-intercept CLPM) have become popular alternatives. Debates about the merits of the CLPM have continued, however, with some researchers arguing that the CLPM is more appropriate than modern alternatives for examining common psychological questions. In this article, I discuss the ways that these defenses of the CLPM fail to acknowledge well-known limitations of the model. I propose some possible sources of confusion regarding these models and provide alternative ways of thinking about the problems with the CLPM. I then show in simulated data that with realistic assumptions, the CLPM is very likely to find spurious cross-lagged effects when they do not exist and can sometimes underestimate these effects when they do exist.

But also...

Testing Prospective Effects in Longitudinal Research: Comparing Seven Competing Cross-Lagged Models

Ulrich Orth
University of Bern

D. Angus Clark
University of Michigan

M. Brent Donnellan
Michigan State University

Richard W. Robins
University of California, Davis

In virtually all areas of psychology, the question of whether a particular construct has a prospective effect on another is of fundamental importance. For decades, the cross-lagged panel model (CLPM) has been the model of choice for addressing this question. However, CLPMs have recently been critiqued, and numerous alternative models have been proposed. Using the association between low self-esteem and depression as a case study, we examined the behavior of seven competing longitudinal models in 10 samples, each with at least four waves of data and sample sizes ranging from 326 to 8,259. The models were compared in terms of convergence, fit statistics, and consistency of parameter estimates. The traditional CLPM and the random intercepts cross-lagged panel model (RI-CLPM) converged in every sample, whereas the other models frequently failed to converge or did not converge properly. The RI-CLPM exhibited better model fit than the CLPM, whereas the CLPM produced more consistent cross-lagged effects (both across and within samples) than the RI-CLPM. We discuss the models from a conceptual perspective, emphasizing that the models test conceptually distinct psychological and developmental processes, and we address the implications of the empirical findings with regard to model selection. Moreover, we provide practical recommendations for researchers interested in testing prospective associations between constructs and suggest using the CLPM when focused on between-person effects and the RI-CLPM when focused on within-person effects.

Keywords: cross-lagged panel models, longitudinal, prospective effects, statistical analyses, structural equation modeling

Supplemental materials: <http://dx.doi.org/10.1037/pspp0000358.supp>

Orth, U., Clark, D. A., Donnellan, M. B., & Robins, R. W. (2021). Testing prospective effects in longitudinal research: Comparing seven competing cross-lagged models. *Journal of Personality and Social Psychology*, 120(4), 1013–1034. <https://doi.org/10.1037/pspp0000358>

But again...

Replicating biased results is only giving us false confidence in erroneous conclusions.

Why Most Cross-Lagged Models Are False

🕒 August 22, 2020 📁 Cross-Lagged Panel Model, Depression, DOI: 10.1037/pspp0000358, Longitudinal Data, Panel Data, Self-Esteem, Spurious Correlations

Abstract

Ulrich Orth, Angus Clark, Brent Donnellan, Richard W. Robins (DOI: 10.1037/pspp0000358) present 10 studies that show the cross-lagged panel model (CLPM) does not fit the data.

This does not stop them from interpreting a statistical artifact of the CLPM as evidence for their vulnerability model of depression. Here I explain in great detail why the CLPM does not fit the data and why it creates an artifactual cross-lagged path from self-esteem to depression. It is sad that the authors, reviewers, and editors were blind to the simple truth that a bad-fitting model should be rejected and that it is unscientific to interpret parameters of models with bad fit. Ignorance of basic scientific principles in a high-profile article reveals poor training and understanding of the scientific method among psychologists. If psychology wants to gain respect and credibility, it needs to take scientific principles more seriously.

Using the CLPM when there are stable between-person differences may lead to

- Spurious relationships
- Undetected relationships
- Relationships in the wrong direction
- A variable being incorrectly identified as causally dominant

The random-intercept cross-lagged panel model

Random-intercept cross-lagged panel model (RI-CLPM)

To extend the CLPM to an RI-CLPM, we need to add two random intercepts and their covariance.

→ CLPM nested under RI-CLPM.

For model estimation, you need at least three waves (in comparison to two waves with the CLPM).

Random-intercept cross-lagged panel model (RI-CLPM)

$$x_{t,i} = c_{x,i} + \mu_{x,t} + \tilde{x}_{t,i}$$

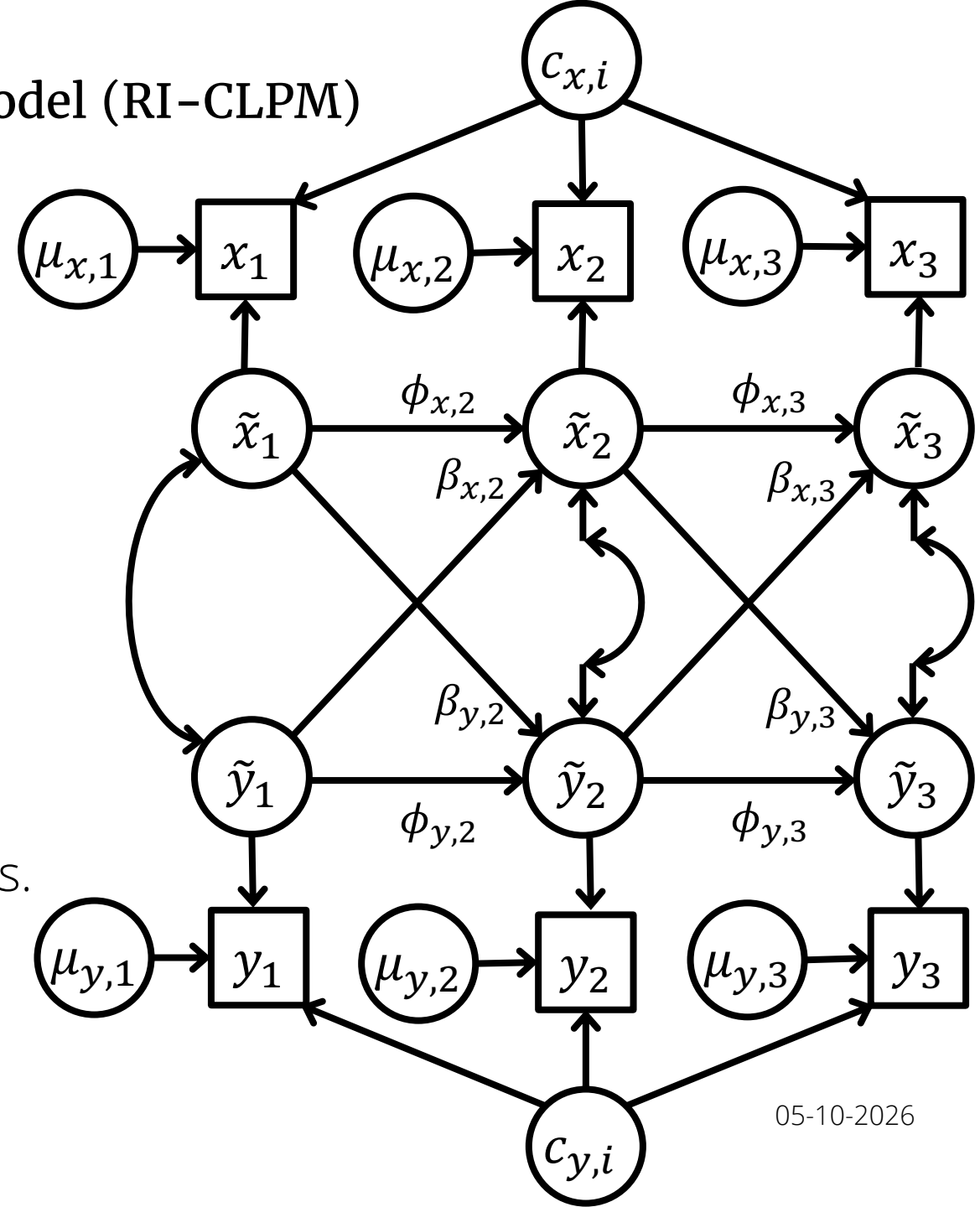
$$\tilde{x}_{t,i} = \phi_{x,t-1} \cdot \tilde{x}_{t-1,i} + \beta_{x,t-1} \cdot \tilde{y}_{t-1,i} + v_{x,t,i}$$

$$y_{t,i} = c_{y,i} + \mu_{y,t} + \tilde{y}_{t,i}$$

$$\tilde{y}_{t,i} = \phi_{y,t-1} \cdot \tilde{y}_{t-1,i} + \beta_{y,t-1} \cdot \tilde{x}_{t-1,i} + v_{y,t,i}$$

Note that $\tilde{x}_{t,i}$ and $\tilde{y}_{t,i}$ now represent the individual's temporal deviations from their expected scores ($c_{x,i} + \mu_{x,t}$).

ϕ and β now represent within-person effects.



Cross-lagged effects β

Reciprocal influences over time within individuals or dyads.

Estimation in R

Using the **lavaan** package, model specification is in **lavaan** code and offers a lot of flexibility.

The **psychonetrics** package also offers RI-CLPM estimation through the **ri_clpm()** function, but is less flexible.

Learn more about RI-CLPM extensions

Mulder, J. D., & Hamaker, E. L. (2021). Three Extensions of the Random Intercept Cross-Lagged Panel Model. *Structural Equation Modeling: A Multidisciplinary Journal*, 28(4), 638–648.
<https://doi.org/10.1080/10705511.2020.1784738>

Extension 1: Including time-invariant predictors and outcomes

Extension 2: Multiple group

Extension 3: Multiple indicators

Nice documentation with **lavaan** code examples:

<https://jeroendmulder.github.io/RI-CLPM/>

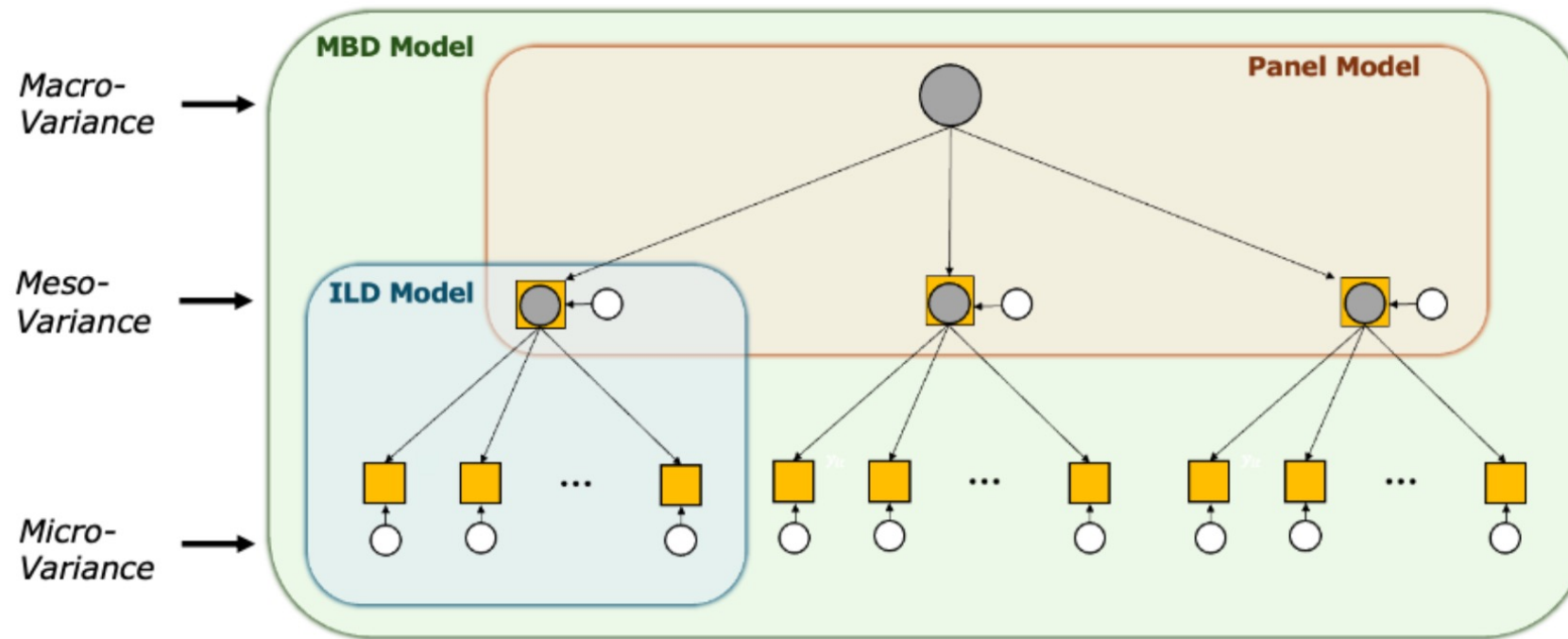
Power analysis for the RI-CLPM: the powRICLPM R-package

Mulder, J. D. (2023). Power Analysis for the Random Intercept Cross-Lagged Panel Model Using the powRICLPM R-Package. *Structural Equation Modeling: A Multidisciplinary Journal*, 30(4), 645–658.
<https://doi.org/10.1080/10705511.2022.2122467>

Nice documentation:

<https://jeroendmulder.github.io/powRICLPM/articles/introduction.html>

Modeling a combination of ILD and panel data



From: Andresen, P. K., Schuurman, N. K., & Hamaker, E. L. (2025, May 23). The Dynamic Measurement-Burst Model: A method for assessing psychological process features at multiple timescales.

https://doi.org/10.31234/osf.io/t4bdf_v1

Only in Mplus



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shaping tomorrow